

Standard Operating Procedure (SOP) EDI 811 Notification Mapping Design Using Altova MapForce

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Based on Existing 810 Notification.mfd Mapping

1. PURPOSE

This document explains how to create the new EDI 811 Notification Mapping (Notification_811.mfd) based on the existing EDI 810 Notification mapping.

Current Requirement:

- Input = Telecom EDI 811 File
 - Output = CSV Notification File
 - Tool = Altova MapForce
 - Automation = FlowForce Server
-

2. CURRENT 810 NOTIFICATION MAPPING ANALYSIS

The existing BGE_Notification.mfd mapping currently performs:

- File name extraction
- String manipulation
- Date formatting
- Amount formatting
- CSV generation
- File path handling
- Validation logic
- Notification output generation

The mapping uses functions like:

Function	Usage
string-length	Length validation
substring	Extract partial values
remove-fileext	Remove file extension
remove-folder	Remove folder path
concat	Build notification values
if-else	Conditional logic
year-from-datetime	Date extraction

These same functions can be reused for EDI 811 Notification mapping.

3. NEW EDI 811 NOTIFICATION MAPPING REQUIREMENT

Input

EDI 811 Telecom File

Example:

```
ST~811~000013004
BIG~20250911~1717924093078
REF~12~1717924093078
HL~1~~1~1
IT1~56~1~M4~55~~SV~0200
SLN~57~~I~1~M4~5~~I
TDS~267188
```

Output

CSV Notification File

Example:

```
FileName,InvoiceNumber,AccountNumber,Amount,Status
ATT_811_001.edi,1717924093078,1717924093078,2671.88,Processed
```

4. RECOMMENDED CSV OUTPUT COLUMNS

CSV Column	EDI Segment Source
FileName	Input File Name
TransactionType	ST01
InvoiceNumber	BIG02
BillingAccount	REF*12
InvoiceDate	BIG01
BillingStartDate	DTM*186
BillingEndDate	DTM*187
ServiceIdentifier	SI Segment
ChargeAmount	AMT/TDS
TotalAmount	TDS
Status	Static Value
ProcessDate	System Date

5. IMPORTANT SEGMENTS REQUIRED FOR NOTIFICATION MAPPING

Segment	Purpose
ST	Transaction Type
BIG	Invoice Information
REF	Billing Account
DTM	Billing Dates
HL	Hierarchy Structure
IT1	Service Billing
SI	Telecom Identifier
SLN	Charge Detail
AMT	Amount
TDS	Total Invoice Amount

6. IMPORTANT MAPFORCE FUNCTIONS REQUIRED

Function	Purpose
remove-fileext	Remove .edi extension
remove-folder	Remove full path
substring	Extract partial values
concat	Build output text
if-else	Validation logic
exists	Null validation
string-length	Value length validation
sum	Amount aggregation
distinct-values	Prevent duplicates
remove-context	Handle nested loops

7. RECOMMENDED MAPFORCE STRUCTURE

Source Side

EDI 811 Schema

Main Segments:

- ISA
- GS
- ST
- BIG
- REF
- DTM
- HL
- IT1
- SI
- SLN
- AMT
- TDS

Target Side

CSV Output Structure

Columns:

- FileName
 - InvoiceNumber
 - AccountNumber
 - ServiceIdentifier
 - Amount
 - Status
-

8. STEP-BY-STEP DEVELOPMENT APPROACH

Step 1 – Copy Existing 810 Notification Mapping

Create new mapping:

BGE_811_Notification.mfd

Reuse:

- File handling logic
 - CSV output logic
 - Date formatting
 - Notification structure
-

Step 2 – Replace 810 Schema with 811 Schema

Import:

X12 811 Version 004010

Step 3 – Map Header Segments

Map:

EDI Segment	CSV Field
BIG02	InvoiceNumber
BIG01	InvoiceDate
REF*12	AccountNumber
TDS	TotalAmount

Step 4 – Add Telecom Segments

New telecom-specific segments required:

Segment	Purpose
HL	Hierarchy processing
SI	Telecom identifiers
SLN	Telecom charges
IT1	Service billing

Step 5 – Build HL Loop Logic

MOST IMPORTANT STEP.

Need:

- Parent-child hierarchy handling
- Nested loop processing
- remove-context() usage

Example:

```
HL
↓
IT1
↓
SLN
```

Step 6 – Add Validation Logic

Need validation for:

- Missing invoice number
 - Missing account number
 - Missing TDS
 - Duplicate records
 - Invalid telecom identifiers
-

Step 7 – Generate CSV Output

Final output:

```
FileName,InvoiceNumber,AccountNumber,Amount,Status  
ATT_811.edi,1717924093078,1717924093078,2671.88,Processed
```

9. FLOWFORCE AUTOMATION

Proposed Process

```
FTP Pickup  
↓  
Local Incoming Folder  
↓  
Run 811 Notification Mapping  
↓  
Generate CSV Output  
↓  
Move CSV to Outgoing Folder  
↓  
Archive EDI File  
↓  
Send Notification
```

10. RECOMMENDED OUTPUT FILE NAME

Recommended mapping name:

BGE_811_Notification.mfd

Recommended CSV name:

ATT_811_Notification.csv

11. MOST IMPORTANT DEVELOPMENT AREAS

Area	Complexity
HL Hierarchy Handling	HIGH
Nested Loop Processing	HIGH
Telecom Segment Handling	HIGH
CSV Generation	MEDIUM
Validation Logic	MEDIUM
Performance Optimization	HIGH

12. FINAL TECHNICAL CONCLUSION

The current 810 Notification mapping can be partially reused for EDI 811 Notification processing.

Reusable logic:

- File handling
- CSV generation
- String formatting
- Date formatting
- Notification structure
- Validation framework

However, EDI 811 requires additional telecom-specific development for:

- HL hierarchy handling
- SI telecom identifiers
- SLN telecom charge detail
- Nested loop processing
- Usage-level processing
- Large file optimization

A separate dedicated mapping is recommended:

`BGE_811_Notification.mfd`

instead of modifying the existing 810 Notification mapping directly.